

DIGITAL FORENSICS CHEAT SHEET

STANDARD: NIST SP 800-86
CLASSIFICATION: LEGAL
EVIDENCE

EVIDENCE ACQUISITION, ANALYSIS & CHAIN OF CUSTODY

01. FUNDAMENTALS & REAL-WORLD CASE BREAKDOWN

Digital Forensics is the application of specialized scientific methods to identify, collect, examine, and analyze cybercrime evidence on digital devices for legal proceedings without tampering with original data.

CASE STUDY: BANK ROBBERY DIGITAL EVIDENCE ANALYSIS

- Laptop:** Digital bank map used for heist planning; photos/videos of past robberies.
- Hard Drive:** Entrance/escape routes; physical security control bypassing blueprints.
- Mobile Phone:** Encrypted chat group participation; call logs related to co-conspirators.
- USB Drive:** Secure evidence collection and laboratory isolation required.

02. NIST FOUR-PHASE INVESTIGATION PROCESS

1. COLLECTION

STAGE 01

Identify and secure physical digital media (PCs, laptops, cameras, USBs). Maintain strict original data integrity and document scene parameters.

2. EXAMINATION

STAGE 02

Filter massive raw datasets to isolate relevant artifacts. Extract specific timeframe media, target user accounts, or specific file extensions.

3. ANALYSIS

STAGE 03

Correlate isolated evidence to draw legal conclusions. Reconstruct events chronologically to demonstrate intent and activity timelines.

4. REPORTING

STAGE 04

Synthesize technical findings into formal reports. Include executive summaries for non-technical stakeholders, management, and court trials.

03. SPECIALIZED BRANCHES OF DIGITAL FORENSICS

COMPUTER

Investigation of desktop PCs and laptops, file systems, and local system artifacts.

MOBILE

Extracting call records, SMS, GPS location history, and app databases from mobile devices.

NETWORK

Analyzing packet captures (PCAPs), traffic flows, and firewalls/IDS logs across infrastructure.

DATABASE

Investigating unauthorized access, SQL injection, row modifications, and data exfiltration.

CLOUD

Examining multi-tenant virtual environments, cloud logs, and remote object storage.

EMAIL

Analyzing email headers, server routing logs, and attachments for phishing or fraud.

04. EVIDENCE ACQUISITION & INTEGRITY PRINCIPLES

Acquiring digital evidence requires adherence to strict legal and operational safeguards. Improper handling compromises data integrity and renders evidence inadmissible in court.

PROPER AUTHORIZATION

Must secure warrants or organizational consent prior to collection. Unapproved collection violates privacy laws and invalidates court standing.

CHAIN OF CUSTODY

Formal paper/digital log recording every transfer, access timestamp, handler identity, and secure storage location to prove tamper-free integrity.

WRITE BLOCKERS

Hardware or software write-blocking devices prevent OS background tasks from altering drive timestamps or data during forensic acquisition.

05. CHAIN OF CUSTODY RECORD STRUCTURE

REQUIRED FIELD	PURPOSE IN COURT	STANDARD OPERATIONAL ENTRY
Evidence Description	Unambiguously identifies physical media.	Seagate 2TB External HDD (S/N: WCX1029384)
Collector Identity	Establishes primary accountability.	Det. J. Miller (#4402) / Lead Forensics Specialist
Date & Time	Establishes timeline of evidence seizure.	2026-09-28 03:15:00 UTC
Storage Location	Verifies physical security compliance.	Evidence Locker B, Safe #04 (Anti-Static Bag)
Access / Handler Log	Tracks every access attempt or transfer.	Transferred to Analyst K. Vance for FTK Imaging

06. WINDOWS FORENSICS: MEMORY VS. DISK IMAGES

DISK IMAGE (Non-Volatile)

HDD/SSD

Bit-by-bit copy of physical storage. Survives reboots. Contains document files, media, browser history, deleted files, system registry, and installed binaries.

MEMORY IMAGE (Volatile)

RAM FIRST

RAM contents lost instantly upon power-off. Holds running processes, open network sockets, decrypted passwords, and unwritten file buffers. **Must acquire FIRST!**

ESSENTIAL FORENSIC TOOLKIT

FTK Imager: GUI tool for creating physical/logical disk images across multiple formats and analyzing file trees.
Autopsy: Open-source GUI platform for disk image analysis, keyword searching, file recovery, and metadata extraction.

DumpIt: Lightweight command-line utility designed for fast, low-footprint RAM memory acquisitions on Windows.
Volatility: Advanced CLI framework for memory analysis using targeted plugins across Windows, Linux, and macOS.